

Design FMEA (qualitative version, based on simulation signals) - VBF-T3R2-DFMEA-01

Qualitative ratings derive from virtual-test risk signals only - this version is annotated "qualitative version, based on simulation signals". Severity (S) and occurrence (O) are three-level (high/medium/low); basis = magnitude of the simulation value's deviation from the threshold. No numbers are written from memory.

#	Failure mode	Failure cause	Simulation signal (detectability basis)	S	O	Design-side mitigation recommendation
1	Lithium plating at negative electrode during fast charge	High charge current density vs anode charge acceptance; insufficient anode kinetics/porosity	cell/r2_p1_4_c_dfn.json:anode_potential_v_min = +0.0364 V - positive in the delivered design (no plating) but margin is only 36 mV above 0	High	Low (in-sim, 45 °C ambient)	Keep anode porosity ≥ 0.40 and $r_{neg} = 1.6 \mu m$; lock 4C current to the re-based nominal (3.32 Ah); re-verify plating margin with production-tolerance electrodes; do NOT displace to low-temperature charge without re-verification
2	Over-temperature at fast charge	I^2R heating vs lumped cooling coefficient $h = 10 \text{ W/m}^2\text{K}$ at 45 °C ambient	cell/r2_p1_4_c_dfn.json:T_max_K = 332.52 K vs 333.15 K limit - passes with 0.63 K margin	High	Medium	Pack-level forced-air/liquid cooling confirmation; larger external surface (cooling area) is a pack DoF; verify T_max under worst-case ambient + fresh-cell high capacity
3	Insufficient capacity against contract	Over-thinning electrodes; manufacturing thickness loss	cell/r2_p1_1_c_dfn.json:capacity_ah = 3.3278 Ah vs $\geq 2 \text{ Ah}$ - 66 % margin	Medium	Low	Hold electrode thickness tolerance bands (min 43 μm pos / 54 μm neg keeps $\geq 2.9 \text{ Ah}$ by scaling); incoming material QC on c_{max} and active fraction
4	Electrolyte oxidative decomposition at high voltage cutoff	Upper voltage 4.2 V with NMC811; electrolyte oxidation window	No DFT endorsement run this case (real_compute = false; honestly skipped) - signal basis absent, flag is knowledge-based caution	Medium	Low (unverified)	Keep upper cutoff 4.2 V; verify electrolyte oxidation stability via true-compute endorsement (run-orca/QE) before physical build; formation protocol per DS §3

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5	Rate collapse on customer-side 5C bursts (regression risk)	Any future particle/porosity regression toward base powder specs	Attribution isolate P2_no_particles (base radii) at 62.5 % retention vs 95 % threshold - sensitivity to particle size is proven by round 2	Medium	Low	Pin particle specification r_pos 1.3 um (± 0.2) / r_neg 1.6 um (± 0.2) as incoming-QC parameter; reject coarser powders at IQC
6	Production process failures (slurry, coating, welding, sealing)	Process-chain defects	No simulation signal (beyond pure-simulation boundary)	N/A	N/A	N/A - complete FMEA including process/supplier failures is beyond the pure-simulation boundary and requires manufacturing data

Conclusion

Highest-risk items: #1 (plating, 36 mV margin) and #2 (fast-charge T_max, 0.63 K margin). Both mitigations are designed into the virtual cell and both signals are verified positive in simulation; the residual margin close to the limits is flagged for pack-level engineering. Full process/supplier FMEA annotated N/A (beyond pure-simulation boundary).